

Time: 2hrs

Max. Marks: 30

Bloom's Taxonomy levels (BL) L1- Remember, L2- understand, L3- apply, L4- Analyse, L5 Evaluate, L6 -Create

Course Outcomes

On successful completion of the course, the students will be able to :

CO1: Interpret modern devices and technologies based on lasers and optical fibres.

CO2 : Explain the basic principles of crystal physics.

CO3 : Summarise the characteristics and applications of superconducting materials, nano materials and smart materials.

CO4 : Illustrate the theory of semiconductors and magnetic materials.

CO5 : Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

PART – A

Attempt all questions

(5 x2 = 10)

Q.No I.		Marks	BL	CO	PO
1	Draw the energy level diagram of the He-Ne laser and mark all the transitions.	2	1	1	1
2	Why do we need a pumping source in a lasing system ?	2	2	1	1
3	Describe the significance of Moseley's law.	2	1	2	1
4	Prove that a bcc structure is more closely packed compared to a sc structure.	2	2	2	1
5	Define miller indices and give two of its significant characteristics.	2	1	2	1

PART – B

Answer any 2 questions from the options provided.

(2 x10 =20)

Q.No.		Marks	BL	CO	PO
II (a)	With the help of suitable diagrams explain the principle, apparatus and working of a three level laser.	7	2	1	2
(b)	Deduce the ratio of spontaneous emission to stimulated emission for a wavelength of 600 nm at 2000 K.	3	3	1	2
	OR				
III (a)	Explain the various characteristics of the different types of optical fibres with suitable diagrams	7	1	1	2
(b)	Numerical aperture of an optical fibre is found to be 0.2441 and the refractive index of the core is 1.5. Calculate the refractive index of the cladding and the acceptance angle.	3	3	1	2

Q. No		Marks	BL	CO	PO
IV (a)	Define bravais lattice, atomic packing fraction and coordination number. Which crystal system has the maximum number of bravais lattice, name them.	7	1	2	2
(b)	Draw the (124), (020) and the (001) crystal planes for the unit cell of a simple cubic system.	3	3	2	2
	OR				
V (a)	Explain the working principle, apparatus and analysis of a Bragg's spectrometer with suitable figures.	7	2	2	2
(b)	Calculate the glancing angle at (110) plane of cubic crystal having axial length 0.26 nm corresponding to the second order. Diffraction maximum for the x-rays of wavelength 0.065 nm.	3	3	2	2

BL1 - 38 %,

BL2 – 31 %

BL3 - 31 %